

Session 2: Classes and Objects Topics: Define class & object, syntax, member functions . Example: Class Note (title, date, is Done).

Tasks

- 1) Create a class called Playlist with properties: name (string), createdOn (date), and isPublic (boolean). Instantiate an object of Playlist and print all its properties**

Code-

```
#include <iostream>
#include <string>
using namespace std;

class Playlist
{
public:
    string name;
    string createdOn;
    bool isPublic;
};

int main()
{
    Playlist p1;

    p1.name = "Tum Hi Ho";
    p1.createdOn = "23-07-2026";
    p1.isPublic = true;

    cout << "Playlist Name : " << p1.name << endl;
    cout << "Created On   : " << p1.createdOn << endl;
    cout << "Public       : " << (p1.isPublic ? "Yes" : "No") << endl;
```

```
    return 0;  
}
```

Output-

Output

```
Playlist Name : Tum Hi Ho  
Created On    : 23-07-2026  
Public        : Yes
```

- 2) Add a member function to your Playlist class named toggle Public() that switches is Public between true and false each time it is called. Demonstrate by toggling the value twice and printing the result each time.**

Code-

```
#include <iostream>
#include <string>
using namespace std;

class Playlist
{
public:
    string name;
    string createdOn;
    bool isPublic;

    void togglePublic()
    {
        isPublic = !isPublic;
    }

    void display()
    {
        cout << "Playlist Name : " << name << endl;
        cout << "Created On   : " << createdOn << endl;
        cout << "Public       : " << (isPublic ? "Yes" : "No") << endl;
        cout << endl;
    }
};

int main()
{
    Playlist p1;

    p1.name = "Tum Hi Ho";
    p1.createdOn = "23-07-2026";
    p1.isPublic = true;
```

```
cout << "Initial Playlist Details:" << endl;
p1.display();

p1.togglePublic();
cout << "After First Toggle:" << endl;
p1.display();

p1.togglePublic();
cout << "After Second Toggle:" << endl;
p1.display();

return 0;
}
```

Output-

```
Output
Initial Playlist Details:
Playlist Name : Tum Hi Ho
Created On    : 23-07-2026
Public       : Yes

After First Toggle:
Playlist Name : Tum Hi Ho
Created On    : 23-07-2026
Public       : No

After Second Toggle:
Playlist Name : Tum Hi Ho
Created On    : 23-07-2026
Public       : Yes
```

3) Create a class called Food Order with properties: order Id (number), restaurant Name (string), and is Delivered (boolean). Write a member function mark Delivered() that sets is Delivered to true and prints a message. Instantiate Food Order and call mark Delivered()

Code-

```
#include <iostream>

#include <string>

using namespace std;

class foodorder
{
public:

    int oderid;

    string name;

    bool isdeliver;

    void markdeliver()
    {
        isdeliver=true;

        cout<<"order has sucsefully deliverd"<<endl;

    }

};
```

```
int main()
{
    foodorder p1;

    p1.oderid= 2;
    p1.name = "Shiv Bhojnalay";
    p1.isdeliver = false;

    cout << "order id : " <<p1.oderid<<endl;
    cout << "restarueant name: " <<p1.name <<endl;
    cout << "is deliver: " <<(p1.isdeliver ? "yes" : "no")<<endl;
    cout << endl;

    p1.markdeliver();

    cout<<"deliverd  :"<<(p1.isdeliver ? "yes" : "no")<<endl;

    return 0;
}
```

Output:

Output

```
order id : 2  
restarueant name: Shiv Bhojnalay  
is deliver: no  
  
order has sucsefully deliverd  
deliverd :yes
```

4) Extend your Playlist class to include a member function add Song(song Title) that adds the song title to an array property called songs. Demonstrate by adding three song titles and displaying the updated songs list.

Hint: Initialize songs as an empty array inside the constructor.

Code-

```
#include <iostream>

#include <string>

#include <vector>

using namespace std;

class Playlist
{
public:
    string name;
    string createdOn;
    bool isPublic;
    vector<string> songs;

    Playlist()
    {
        songs = {};
    }

    void addSong(string songTitle)
```



```
{
    songs.push_back(songTitle);
}

void displaySongs()
{
    cout << "Playlist Name : " << name << endl;
    cout << "Songs List:" << endl;

    for (int i = 0; i < songs.size(); i++)
    {
        cout << i + 1 << ". " << songs[i] << endl;
    }
}

};

int main()
{
    Playlist p1;

    p1.name = "My Playlist";
    p1.createdOn = "23-07-2026";
    p1.isPublic = true;

    // Add three songs
    p1.addSong("Tum Hi Ho");
```

```
p1.addSong("Kesariya");  
p1.addSong("Apna Bana Le");
```

```
p1.displaySongs();
```

```
return 0;
```

```
}
```

Output:

Output

Playlist Name : My Playlist

Songs List:

1. Tum Hi Ho

2. Kesariya

3. Apna Bana Le

5) Refactor your Food Order class so that the constructor takes an object as a parameter (with keys order Id, restaurant Name, is Delivered) instead of separate arguments. Update your instantiation code to use this new constructor signature

Code-

```
#include <iostream>
#include <string>
using namespace std;

struct FoodOrderData {
    int orderId;
    string restaurantName;
    bool isDelivered;
};

class FoodOrder {
public:
    int orderId;
    string restaurantName;
    bool isDelivered;

    FoodOrder(FoodOrderData data) {
        orderId = data.orderId;
        restaurantName = data.restaurantName;
        isDelivered = data.isDelivered;
    }

    void display() {
        cout << "Order ID: " << orderId << endl;
        cout << "Restaurant: " << restaurantName << endl;
        cout << "Delivered: " << (isDelivered ? "Yes" : "No") << endl;
    }
};

int main() {
```

```
FoodOrderData order = {101, "Domino's Pizza", true};

FoodOrder foodOrder(order);

foodOrder.display();

return 0;
}
```

Output:

```
Output
Order ID: 101
Restaurant: Domino's Pizza
Delivered: Yes
```

